



Shorya Sethia
Engineering Physics
Indian Institute of Technology Bombay

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B.Tech.
Gender: Male
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Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2026	
Intermediate	CBSE	Global Public School	2022	
Matriculation	CBSE	Global Public School	2020	

SCHOLASTIC ACHIEVEMENTS

- Secured **99.25** All India Percentile in the **JEE Mains** examination among over **1.02 million+** candidates (2022)
- Accomplished a **Top 3%** national rank in **JEE Advanced**, competing against **0.155 million+** candidates (2022)
- Awarded **Branch Change (Top 7%)** in freshmen year owing to exemplary overall academic performance (2023)
- Ranked in **Top 100** out of **0.77 million+** students in **CBSE Aryabhatta Ganit Challenge** (Ajmer Region) (2019)
- Conferred **INSPIRE** scholarship worth **INR 60K** by **GOI** for **top 1%** in **1.4 million+** in CBSE board (2022)
- Grabbed **Top** grades (**AA/AB**) in **13+** courses at IIT Bombay over 6 semesters | **GSSoC** Project Admin (2025)

PROFESSIONAL EXPERIENCE

Machine Learning Research Intern | University of South Carolina - USA (Sept'24 - Present)
Under the guidance of **Prof. Dr. Amitava Das**, Artificial Intelligence Institute, University of South Carolina (AIISC)

- Investigating **Safety Alignment & Adversarial Robustness** of **Text-to-Image** models through training-free model editing and fine-tuning methods (LoRA, Direct Preference Optimization (**DPO**) and Speculative Decoding)
- Created novel **10K image-prompt pairs benchmark** (safe/attacked prompts) for **Query-Free** attacks manipulating **text encoder (CLIP)** without querying the model, designing **evaluation** metrics inspired by **Pick-a-Pic**

Data Analyst | NoQs Digital (June'24 - July'24)
Received **Letter of Recommendation** and **STAR Intern Award** for exceptional performance from Founder & CEO

- Developed dynamic **Dashboards** using **SQL**, **PowerBI** & **Google Sheets** queries to drive data-centric decisions
- Built web app with **CRUD** capabilities using **Google Apps Script** and Sheets backend managing **3K+** records
- Performed large-scale **data mining**, competitor filtering on **10K+** entries to support **M&A** acquisition strategy

PUBLICATIONS

Markdrop | PyPI - Python Package Index | Documents to Markdown and HTML (Dec'24 - Present)
Open-Source Self-Project with 15,000+ installs and 140+ stars. Available using pip install markdrop

- Engineered & automated **PDF-to-Markdown** conversion pipeline using **Table Transformers**, **Docling**, **XRef IDs** for structuring **textual & non-textual** data; working on optimized **RAG** with semantic caching & reranking
- Developed** method for generating **in-place** self-explanatory descriptions for non-textual content using **5+ LLMs**
- Enabled modular configuration setup, **CLI** & **Docker** based **deployment**, added one-click table download as **Excel**

KEY TECHNICAL PROJECTS

BackTube | Backend for YouTube and Twitter like Platform | Self-Project (Jun'25 - Present)

- Built **Express microservice** with **routes**, **controllers** and **middlewares** & tested all endpoints using **Postman**
- Designed data models via **Mongoose**, wrote simple **MongoDB** aggregation pipelines. Implemented **JWT** & **cookie** based **authentication** with refresh/access tokens, bcrypt **password hashing** & secure middleware routing
- Developed two-step media upload using **Multer** & **Cloudinary** with central response, validation, and error handling

Blogify | A Full Stack Blogging Platform | Seasons of Code | Web and Coding Club (May'23 - Aug'23)

- Developed a blog platform with **React**, **Redux Toolkit**, **React hooks** and **Appwrite** (Backend as a service) featuring user authentication, saving to **local storage** enhancing improved user experience and **slug-based routing**
- Implemented **TinyMCE** rich text editor with **Tailwind CSS** for responsive design, role-based access control, **SEO-optimized** URL structure. Deployed on **Vercel** with automated **CI/CD** pipeline, performance optimization

AgentCode | Agentic Web-Based IDE | Seasons of Code | Web and Coding Club (May'25 - Jul'25)

- Developed web IDE with **React**, **Tailwind CSS**, **Monaco** & **Pyodide** for in-browser live python code execution
- Built **Planner**, **Developer** & **Orchestrator** agents via **LangGraph**, architected backend using **FastAPI** server
- Added **AST-based** internal search, **Merkle Trees** to track changes, used **Tavily** retrieval for external references

Emoji-Aware Tweet Sentiment Classification | Self-Project (Aug'25)

- Processed **1.6Mn** tweets with **tokenization**, **stemming**, **lemmatization** & **sentiment mapping** for **960+** emojis
- Vectorized text using **TF-IDF (50K dim)** with **N-Gram** features; combined with emoji scores using weighted score
- Trained **Logistic Regression** using **GridSearch**, **Cross Validation**; achieved **78.49%** accuracy, **0.79 F1 Score**

OTHER PROJECTS AND COMPETITIONS

Rail Planner | Self-Project | Data Structures & Algorithms | **LeetCode: 850+ Qs, Rating: 1724** (May'25)

- Implemented **Trie** + **DFS** based **autocomplete** for station names and **KMP** for efficient keyword search in reviews
- Used **max heap** to filter reviews by rating. Applied **Merge Sort** on **linked lists** to sort journeys/reviews efficiently
- Modeled stations network as a **Graph** and stored journeys with **Binary Search Trees** for fast lookup and updates

Image Caption Generation | IIT Guwahati | Consulting and Analytics Club (May'24 - Jun'24)

- Build image captioning model using **pre-trained VGG16** as **encoder** and network of **LSTM** layers as the **decoder**
- Fine-tuned the model on the **Flickr8K** dataset; achieved **BLEU-1** & **BLEU-2** scores of **0.57** & **0.34** respectively

SimpleChat & SimpleFTP | Socket Programming | Computer Networks | Self-Project (Jun'25)

- Engineered a **TCP** based **bidirectional** chat app in **C++** using **Winsock** API, achieving full-duplex communication
- Implemented binary-safe file transfer over **TCP**, following **bind-listen-accept** cycle and **three-way handshake**

Autonomous Sun Tracker | Course Project - PH222 | Guide: Prof. Maniraj Mahalingam (Mar'24)

- Engineered **Arduino-C++** based dual-axis sun tracking system using **4 LDR** sensors, **bi-directional** servo control
- Implemented **algorithm** for a **line-follower** bot equipped with solar panel to **scan, log & compare** solar voltages over a **2D** sweep, autonomously repositioning itself & panel to the **maximum** irradiance orientation in next cycle

TrueMeasure | Course Project - DH803 | Guide: Prof. Nirmal Punjabi (Aug'24 - Nov'24)

- Modeled respiratory rate prediction for **wearable smartwatches** by extracting features from ECG and PPG signals
- Achieved **90.44% R² Score** using **Random Forest**, outperforming **XGBoost (81.78%)**, **LightGBM (56.98%)**
- Developed 5-class ECG arrhythmia classifier with **97.98%** accuracy using weighted **ensemble** of **LSTM** and **GRU**

Deep Generative Models | Seasons of Code | Web and Coding Club (May'24 - Jul'24)

- Trained **GAN** on **200K+ CelebA** faces & **Conditional GAN** on **MNIST** for **label-conditioned** image synthesis
- Investigated & compared multiple **VAE** architectures to explore their effectiveness in learning **latent representation**
- Tricked **CNN** to **misclassify** by **FGSM**, defended via **denoising autoencoder**, **Block Switching** of **4 VGG16s**

Qualcomm VisionX | Ranked 4 (100+ teams) & Won **INR 25.5K** | Techfest, IIT B (Nov'24 - Dec'24)

- Implemented **Reflection Removal** using transformer with cross-scale attention and multi-scale fusion, boosting **PSNR** from 19.23 to 23.61, **SSIM** to 0.83, designed **triple-loss** optimization for transformer-based feature extraction

COURSES UNDERTAKEN

CS & DS	Computer Programming & Utilization, Data Structures and Algorithms, AI and Data Science, Logic for Computer Science, DevOps, Introduction to MCP, Multi Agent Systems, Discrete Structures
Maths	Calculus, Linear Algebra, Differential Equations, Complex Analysis, Numerical Analysis, Statistics

POSITIONS OF RESPONSIBILITY

Operations Coordinator | Mood Indigo, IIT Bombay (May'23 - Jan'24)

Asia's Largest College Cultural Festival | Footfall: 1,46,000+ | Colleges: 4000+ | Events: 240+ | Viewership: 3+ hours

- Worked in team of **55** members, responsible for smooth execution of events attended by a crowd of **140K+** people
- Responsible** for **0.45 Mn+ Diesel Generator deals**, assistance in **0.13 Mn+ High-Capacity Crane Contract**
- Lead** the security and crowd management for the safe conduction of **International Music Festival & concerts**

Mentor | IIT Bombay (Aug'23 - Jul'25)

- Winter in Data Science (Analytics Club)** | Led **9** students through **CNN**, **Autoencoders**, **GAN**, & **Diffusion**
- Seasons of Codes (WnCC)** | Guided **10** folks in **Transfer** and **Ensemble Learning** using Inception and ResNet
- Code Wars (WnCC)** | Mentored **6** freshmen in **Python**, **Git**, and **GitHub** through team-based coding projects
- FinSearch (Finance Club)** | Mentored **16** students in **Option Pricing** & **MLP/RNN** forecast using log returns

TECHNICAL SKILLS

Languages	C++, Python, JavaScript, Bash, SQL
Development	React, TypeScript, Tailwind, Express, Node.js, Django, Flask, FastAPI, Nginx
CS	Data Structures, Computer Networks, DBMS, Operating System, OOPS, System Design
Data Science	NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, TensorFlow, PyTorch, NLTK
Softwares	Git, GitHub, Docker Desktop, MongoDB Compass, Postman, MySQL, L ^A T _E X AutoCAD

EXTRACURRICULAR ACTIVITIES

- Eligible for **INR 0.15 million** equity-free grant from DSSE, IIT Bombay **IDEAS** Program (L1C15) for **OneMail.ai**
- Led **Mood Indigo's Mumbai multicity** with **1000+** participants in over **20+** competitions & events (Dec'23)
- Built financial Q&A **data pipeline** from annual reports at **AfterQuery (YC X25)**, earning **INR 39K** (May'25)
- Completed **McKinsey Forward Program** on **problem-solving** & adaptability, awarded global certificate (Jul'25)
- Completed **Goldman Sachs Virtual Experience**, tackling **10+** tasks strengthening analytical & problem-solving
- Worked with **NSS, IIT Bombay** under **Sustainable Social Development** on **Cloth Collection Campaign**